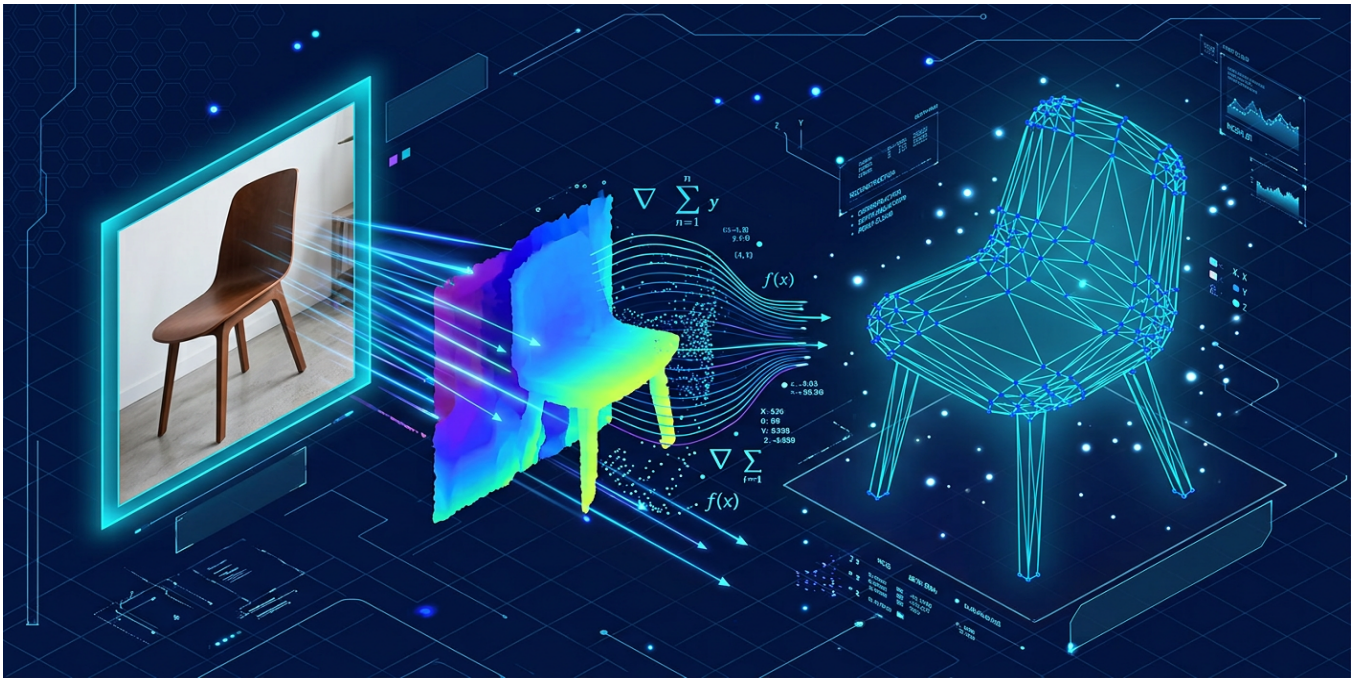
Image-to-3D Construction

Website-Based Single-Image 3D Reconstruction & Generation Platform

Course Project — 3D Computer Vision | Prepared: November 2026



1. Project Description (Creative Brief)

Vision: What if any photo could become a real, rotatable 3D object in seconds? Our project — **"PixelForge 3D"** — is a web application where a user simply logs in, uploads a single 2D image (a product photo, a toy, a chair, a sneaker), and the system reconstructs a fully textured **3D construction** of that object: a downloadable mesh (OBJ/GLB) plus a live 3D viewer in the browser. Behind the upload button sits the heart of 3D computer vision: monocular depth estimation, multi-view diffusion, large reconstruction models (LRMs), NeRF / 3D Gaussian Splatting, and mesh extraction.

The website is the front door; the 3D pipeline is the engine. A user account stores their uploads and generated models in a gallery, can preview the 3D result with orbit controls (three.js / model-viewer), spin a 360° turntable video, and download the asset for Blender, Unity, or 3D printing.

2. Objectives

- Build a login-secured web app with image upload, job queue, and 3D model gallery.
- Integrate a production-grade image-to-3D model (e.g., TripoSR, InstantMesh, Hunyuan3D-2, or TRELLIS) as the reconstruction backend.
- Render results in-browser with a real-time viewer (three.js, model-viewer, or a Gaussian-splat viewer).
- Demonstrate the underlying 3D computer vision concepts: depth, camera pose, multi-view geometry, neural radiance fields / splatting.
- Allow export of standard 3D formats (OBJ, GLB/GLTF, PLY) for downstream use.

3. System Architecture (What We Will Build)

Layer	Components	Tech Choices
Frontend website	Login/signup, upload UI, 3D viewer, gallery, download buttons	React/Next.js or Flask templates; three.js / <model-viewer>
Backend API	Auth (JWT), upload handling, async job queue, result storage	FastAPI/Flask + Celery/RQ + PostgreSQL + S3/local storage
3D engine	Background removal, image-to-3D inference, mesh/PLY export, turntable render	TripoSR / InstantMesh / Hunyuan3D-2 / TRELLIS (ONNX/PyTorch, GPU)
Viewer	Orbit 3D view + 360° video in browser	three.js GLTFLoader, PlayCanvas, or gsplat viewer

4. Pipeline Steps (Course Scope)

- **Step 1 — Input:** user logs in and uploads one image (JPG/PNG).
- **Step 2 — Preprocess:** background removal (rembg/BiRefNet), re-centering, resizing.
- **Step 3 — Reconstruct:** feed-forward image-to-3D model produces a triplane/NeRF/Gaussian/mesh representation.
- **Step 4 — Extract:** convert to textured mesh (marching cubes / FlexiCubes) or keep as 3D Gaussians (.ply).
- **Step 5 — Present:** show in the web viewer, render turntable video, enable OBJ/GLB download.
- **Step 6 — Evaluate:** compare against references (PSNR/SSIM/LPIPS, Chamfer distance) and document limitations.

5. Reference GitHub Repositories — Image-to-3D, Ranked by Stars

All links are direct, publicly accessible GitHub repositories, verified November 2026 and sorted by GitHub star count (highest first). Stars are approximate live counts. "Web demo" lists the project/demo website where available. These are the production-grade, fully functional repos best matching our goal of a website that turns an uploaded image into a 3D construction.

#	Repository (direct GitHub link)	Stars	Why it matches our project	Website / live demo
1	**3D Gaussian Splatting** https://github.com/graphdeco-inria/gaussian-splatting	~24.0k	Reference real-time radiance-field renderer; the foundation for modern image/scene-to-3D output and in-browser 3D viewing.	https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/
2	**Instant-NGP** https://github.com/NVlabs/instant-ngp	~17.6k	NVIDIA's lightning-fast NeRF/SDF engine used as the 3D backbone in many image-to-3D pipelines (e.g., stable-dreamfusion).	https://www.nvidia.com/en-us/research/
3	**Hunyuan3D-2** https://github.com/Tencent-Hunyuan/Hunyuan3D-2	~15.3k	Tencent's production image-to-3D system: single image → high-res textured 3D asset (GLB/OBJ). Best overall quality; hosted platform available.	https://3d.hunyuan.tencent.com/
4	**TRELLIS** https://github.com/microsoft/TRELLIS	~14.2k	Microsoft's structured-latent image/text-to-3D model (CVPR'25); outputs mesh, Gaussians, or radiance fields; official Hugging Face demo.	https://trellis3d.github.io/
5	**nerfstudio** https://github.com/nerfstudio-project/nerfstudio	~12.1k	Modular NeRF/3DGS studio with a real-time web viewer — ideal reference for the website + 3D viewer part of our project.	https://nerf.studio/
6	**Shap-E** https://github.com/openai/shap-e	~12.3k	OpenAI's image/text-to-3D implicit model with ready image-to-3D notebooks; simple API-style reference implementation.	https://openai.com/research/shap-e
7	**TRELLIS.2** https://github.com/microsoft/TRELLIS.2	~12.7k	4B-parameter successor for high-fidelity image-to-3D with full PBR materials and arbitrary topology.	https://microsoft.github.io/TRELLIS.2/
8	**stable-dreamfusion** https://github.com/ashawkey/stable-dreamfusion	~8.9k	Text & image-to-3D with NeRF + diffusion (Zero-1-to-3 backend) and mesh export — classic course-project reference.	—
9	**threestudio** https://github.com/threestudio-project/threestudio	~7.1k	Unified framework for image-to-3D generation (Stable Zero123 etc.) with many ready configs; great for experiments.	https://threestudio.github.io/
10	**TripoSR** https://github.com/VAST-AI-Research/TripoSR	~7.3k	Fastest production single-image→3D feed-forward model (<0.5 s on A100, runs on ~6 GB GPU); outputs OBJ/GLB; Hugging Face demo.	https://huggingface.co/spaces/stabilityai/TripoSR
11	**Wonder3D** https://github.com/xxlong0/Wonder3D	~5.5k	Single image → consistent multi-view normals + fused textured mesh in 2–3 min (CVPR'24 Highlight); HF demo + Colab.	https://www.xxlong.site/Wonder3D/
12	**InstantMesh** https://github.com/TencentARC/InstantMesh	~4.6k	Sparse-view large reconstruction model: clean 3D mesh from one image in ~1 min; Apache-2.0; CLI + Gradio demos.	https://huggingface.co/spaces/TencentARC/InstantMesh
13	**DreamGaussian** https://github.com/dreamgaussian/dreamgaussian	~4.4k	Image/text-to-3D via generative Gaussian splatting + mesh refinement; Colab & Gradio demos; fast (~minutes).	https://dreamgaussian.github.io/
14	**Unique3D** https://github.com/AiuniAI/Unique3D	~3.6k	High-quality textured mesh from a single wild image in ~30 s (NeurIPS'24); project page, HF demo, online demo.	https://unique3d.github.io/
15	**Zero-1-to-3 (zero123)** https://github.com/cvlab-columbia/zero123	~3.1k	The pioneering zero-shot one-image-to-3D diffusion model (ICCV'23) that powers many later pipelines.	https://zero123.cs.columbia.edu/

#	Repository (direct GitHub link)	Stars	Why it matches our project	Website / live demo
16	**LGM** https://github.com/3DTopia/LGM	~2.1k	Large multi-view Gaussian model: text/image → high-res 3D Gaussians → mesh (ECCV'24 Oral); real-time rendering.	https://3dtopia.github.io/lgm/
17	**Zero123++** https://github.com/SUDO-AI-3D/zero123plus	~2.1k	Single image → consistent multi-view diffusion; used inside InstantMesh for white-background view generation.	https://sudo-ai-3d.github.io/zero123plus/
18	**PartCrafter** https://github.com/wgsxm/PartCrafter	~2.5k	Single RGB image → structured, part-based 3D mesh via compositional latent diffusion (NeurIPS'25).	https://wgsxm.github.io/PartCrafter/
19	**Make-It-3D** https://github.com/junshutang/Make-It-3D	~1.9k	High-fidelity 3D creation from a single image with diffusion prior (ICCV'23).	https://make-it-3d.github.io/
20	**One-2-3-45** https://github.com/One-2-3-45/One-2-3-45	~1.7k	Any single image → 3D mesh in ~45 s without per-shape optimization (NeurIPS'23); Gradio APIs + Docker demo.	https://one-2-3-45.github.io/
21	**Magic123** https://github.com/guochengqian/Magic123	~1.6k	One image → high-quality 3D object using both 2D and 3D diffusion priors (ICLR'24); mesh export included.	https://guochengqian.github.io/project/magic123/
22	**TripoSplat** https://github.com/VAST-AI-Research/TripoSplat	~1.4k	Single image → variable-count 3D Gaussians; tiny inference-only codebase (~2k LOC), near-zero dependencies.	https://huggingface.co/spaces/VAST-AI/TripoSplat
23	**OpenLRM** https://github.com/3DTopia/OpenLRM	~1.3k	Open-source Large Reconstruction Model: image-to-3D in ~10 s on A100; Gradio app included in repo.	https://huggingface.co/spaces/zxhezexin/OpenLRM
24	**Hunyuan3D-2.1** https://github.com/Tencent-Hunyuan/Hunyuan3D-2.1	~4.3k	Image→3D with production-ready PBR materials; lighter successor variant of Hunyuan3D-2.	https://3d.hunyuan.tencent.com/

6. Suggested Stack for Our Build

Recommended primary backend: TripoSR or InstantMesh (fast, light, Apache-2.0/MIT, easy to wrap in a web API). **Highest quality option:** Hunyuan3D-2 / TRELLIS (needs more VRAM). **Viewer:** three.js GLTFLoader for GLB meshes or a splat viewer for .ply output, embedded in a simple login-protected site (Flask/FastAPI + React or server-rendered templates).

Note: If you share the official course syllabus, I can align the milestones, evaluation criteria, and grading sections of this document with it exactly.